

Enhancing Group Attention for Off-Road Semantic Segmentation via Transition-Aware Refinement

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Abstract—We propose EGFormer, a semantic segmentation framework for robust terrain understanding in off-road autonomous driving. In unstructured environments, subtle transition zones between terrain types critically influence drivability, making reliable modeling of these regions essential. Building upon a group-attention-based segmentation backbone, EGFormer introduces a transition-aware refinement mechanism that explicitly enhances feature representations in terrain transition bands, enabling more reliable segmentation of gradually blending surfaces. Experiments on the RUGD and RELLIS-3D datasets show that while overall mean Intersection over Union (mIoU) improves modestly, boundary IoU (bIoU), which more directly reflects prediction quality near mixed terrain transitions, achieves more substantial gains. Under data-limited training conditions, EGFormer exhibits reduced performance degradation compared to baseline methods, indicating improved robustness. Qualitative results further demonstrate more stable and semantically consistent predictions in ambiguous regions, even under noisy or incomplete annotations. These results suggest that transition-focused feature refinement improves reliability in the ambiguous transition regions most critical to off-road terrain understanding.

Index Terms—Off-road segmentation, semantic segmentation, transition-band modeling, edge refinement.

I. INTRODUCTION

OFF-ROAD autonomous driving requires reliable terrain perception under highly unstructured and ambiguous conditions. Unlike urban roads, off-road environments contain heterogeneous surface materials that often form gradual and visually ambiguous transitions rather than clearly separable regions [1]. These transition zones directly affect drivability and safety, making accurate terrain understanding essential for autonomous navigation. Although semantic segmentation provides pixel-level terrain labels, off-road boundaries are often ill-defined, and visually similar regions may correspond to substantially different physical properties. Small segmentation errors in such mixed regions can propagate to unsafe decisions. Recent studies have explored drivability estimation and semantic segmentation for off-road perception [2], [3], [4], [5], [6], but

most methods mainly optimize region-level accuracy or global consistency. As a result, subtle feature variations and uncertainty within terrain transition bands remain under-modeled.

This limitation is particularly evident in encoder-decoder frameworks, where boundary and transition regions are learned only implicitly from dense supervision. Moreover, off-road datasets often contain annotation noise and ambiguous class boundaries, violating the assumption of uniform label reliability [7] and biasing models toward overconfident predictions in uncertain regions.

From a signal processing perspective [8], transition regions exhibit weak alignment between local appearance cues, spatial continuity, and semantic identity, leading to high intra-class variance and low inter-class separability. Recent signal-processing-oriented semantic segmentation studies have addressed representation ambiguity through frequency-guided denoising [9], position-aware differential denoising [10], and signal-processing-based RGB-D feature fusion [11]. In contrast, EGFormer neither operates in the frequency domain nor relies on multi-modal inputs; instead, it directly models semantically ambiguous transition bands from group-level labels. This is more suitable for off-road perception, where gradual terrain mixing and annotation uncertainty are the primary challenge, rather than thin contours or high-frequency spectral noise.

To address these challenges, we propose EGFormer, a transition-aware semantic segmentation framework for off-road perception. EGFormer introduces an Edge Refinement (ER) module that explicitly enhances high-uncertainty transition bands rather than uniformly sharpening narrow contours. By selectively reinforcing local consistency near terrain transitions while preserving efficient global reasoning, the proposed framework improves boundary robustness with minimal computational overhead, making it suitable for resource-constrained off-road robotic platforms.

The main contributions of this work are summarized as follows:

- We propose an Edge Refinement module for transition-aware refinement in semantically ambiguous terrain bands.
- The proposed framework improves robustness in ambiguous transition regions with minimal computational overhead.
- EGFormer achieves competitive mIoU and improved transition-region quality on RUGD and RELLIS-3D.

II. METHODOLOGY

We propose EGFormer, a parameter-efficient transition-aware refinement framework built upon GA-Nav [5] for robust off-road semantic segmentation. By retaining group attention for efficient region-level reasoning and introducing an Edge

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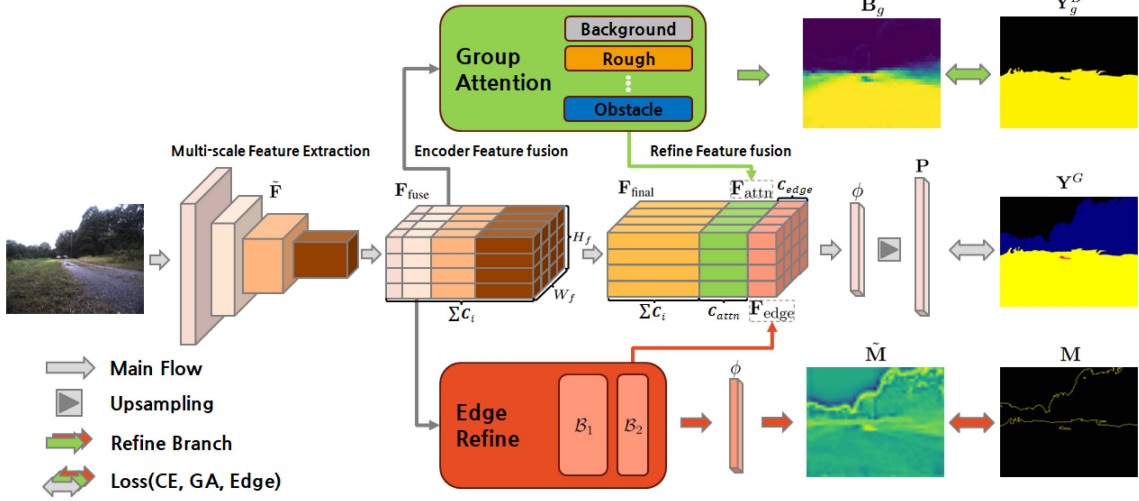


Fig. 1. **EGFormer Network Architecture.** Overview of the proposed EGFormer framework, which incorporates an Edge Refinement (ER) branch as a transition-aware refinement module to explicitly enhance feature representations in terrain transition regions while preserving efficient group-attention-based semantic reasoning.

Refinement (ER) branch, EGFormer explicitly enhances feature representations in high-uncertainty terrain transition regions that are particularly critical to safe off-road navigation. Unlike conventional boundary-focused refinement that mainly sharpens narrow contours, EGFormer selectively focuses on semantically ambiguous transition bands.

A. Task Formulation and Group Representation

Given an RGB image $\mathbf{I} \in \mathbb{R}^{3 \times H \times W}$, the goal is to predict a group-level semantic label map $\hat{\mathbf{Y}}^G \in \mathbb{Z}^{H \times W}$, where each pixel belongs to one of G terrain groups defined by drivability characteristics. Following GA-Nav, fine-grained semantic classes are mapped to group-level labels to reduce class imbalance and stabilize training in unstructured off-road environments. This grouping strategy simplifies the learning objective while retaining task-relevant semantic distinctions, allowing the network to focus on differences that directly affect vehicle traversability and planning decisions.

B. Transition-Band Edge Mask

To explicitly supervise terrain transitions, we derive a binary edge mask $\mathbf{M} \in \{0, 1\}^{H \times W}$ from the group-level ground truth labels $\mathbf{Y}^G$. A pixel is considered part of a transition band if any of its four-connected neighbors belongs to a different terrain group:

$$M(p) = \mathbb{I}[\exists d \in \mathcal{N}_4 : Y^G(p) \neq Y^G(p + d)], \quad (1)$$

where $p = (h, w)$ and $\mathcal{N}_4$ denotes the four-connected neighborhood.

Unlike gradient-based edge detectors that respond to low-level appearance changes, this formulation captures *semantic discontinuities* between terrain groups. Importantly, it represents transition regions as spatially extended bands rather than thin contours, reflecting the gradual and mixed nature of off-road terrain boundaries. By explicitly modeling these ambiguous regions, the proposed edge mask provides targeted supervision where segmentation uncertainty is highest.

C. Edge Refinement Module

As shown in Fig. 1, the ER module operates on the fused decoder feature $\mathbf{F}_{\text{fuse}}$, which aggregates multi-scale contextual information from the backbone and group attention. The ER branch consists of lightweight convolutional layers followed by a 1×1 projection to produce a transition-aware feature representation:

$$\mathbf{F}_{\text{edge}} = \phi_{\text{ER}}(\mathbf{F}_{\text{fuse}}). \quad (2)$$

The ER branch predicts an edge probability map $\tilde{\mathbf{M}}$, supervised by the transition-band edge mask $\mathbf{M}$. The resulting transition-aware feature $\mathbf{F}_{\text{edge}}$ is concatenated with the group-attention feature and the original fused feature to form the final segmentation representation. By explicitly injecting transition-focused features into the prediction pathway, ER guides the decoder to allocate additional representational capacity to ambiguous regions, improving prediction stability near class boundaries without uniformly sharpening all contours.

D. Training Objective

The edge refinement branch is trained using a weighted binary cross-entropy loss:

$$L_{\text{edge}} = -\alpha \mathbf{M} \log \tilde{\mathbf{M}} - (1 - \mathbf{M}) \log(1 - \tilde{\mathbf{M}}), \quad (3)$$

where α balances the contribution of transition-band pixels. This weighting mitigates the class imbalance between edge and non-edge regions [7] and prevents the model from over-emphasizing trivial background areas.

The overall training objective is defined as

$$L = L_{\text{CE}} + \lambda_{\text{GA}} L_{\text{GA}} + \lambda_{\text{edge}} L_{\text{edge}}, \quad (4)$$

where L_{CE} is the group-level segmentation loss and L_{GA} is adopted from GA-Nav. We set $\lambda_{\text{GA}} = 1.0$ following the original GA-Nav formulation and choose $\lambda_{\text{edge}} = 0.5$ empirically, as preliminary comparisons with $\lambda_{\text{edge}} \in \{0.1, 0.5, 1.0\}$ showed

TABLE I

COMPARISON ON RUGD AND RELIS-3D. WE REPORT PER-GROUP IOU (%), MEAN IOU (mIoU $\uparrow$), AND AVERAGE ACCURACY (aAcc $\uparrow$). ASTERISKS (*) DENOTE TRANSFORMER-BASED METHODS. BEST HIGHLIGHTED IN BOLD/UNDERLINE AS APPROPRIATE **BOLD**, SECOND BEST UNDERLINED

Dataset	Methods (IoU)	Smooth	Rough	Bumpy	Forbidden	Obstacle	Background	mIoU $\uparrow$	aAcc $\uparrow$
RUGD	DeepLabv3+ [12]	5.86	84.99	50.40	25.04	87.50	81.47	55.88	91.51
	DANet [13]	2.26	81.47	8.69	15.00	82.54	74.86	44.14	88.81
	OCRNet [14]	66.29	89.47	76.15	59.14	88.77	79.17	76.50	93.46
	BiSeNetv2 [15]	24.27	89.99	<u>89.99</u>	83.31	90.93	75.29	75.10	93.40
	CGNet [16]	40.84	90.39	85.67	76.21	89.75	74.48	76.22	93.29
	FastSCNN [17]	83.03	92.82	87.69	81.05	90.94	75.11	85.11	94.77
	*SETR [18]	89.77	92.46	84.58	70.33	89.55	70.47	82.86	94.09
	*DPT [19]	1.04	81.23	22.98	25.84	89.18	74.50	49.13	88.77
	*SegFormer [20]	93.26	93.16	87.56	77.31	91.20	78.50	86.83	95.17
	*SegNeXt [21]	90.39	91.17	83.96	65.43	87.80	68.17	81.15	93.22
	*GA-Nav [5]	<u>95.15</u>	<u>94.45</u>	89.83	<u>86.25</u>	<u>91.95</u>	76.86	<u>89.08</u>	<u>95.66</u>
	*EGFormer (ours)	95.18	94.51	90.00	86.75	92.06	<u>76.99</u>	89.25	95.78
RELLIS-3D	DeepLabv3+ [12]	65.76	79.84	19.72	47.52	64.88	95.92	62.27	85.84
	DANet [13]	72.93	85.18	13.10	60.60	70.53	95.65	66.38	89.11
	OCRNet [14]	74.67	83.04	27.76	60.44	62.35	92.58	66.81	86.95
	BiSeNetv2 [15]	65.56	73.24	39.35	48.17	71.91	93.78	65.33	83.03
	CGNet [16]	62.84	74.17	49.57	45.41	68.88	94.53	65.90	82.70
	FastSCNN [17]	67.06	77.60	56.49	49.76	70.31	94.43	69.27	84.51
	*SETR [18]	65.37	78.64	40.89	52.59	63.80	91.87	65.53	83.59
	*DPT [19]	5.42	76.65	47.13	54.87	62.74	85.50	55.38	81.61
	*SegFormer [20]	60.28	79.78	<u>53.35</u>	53.78	70.15	94.37	68.62	85.37
	*SegNeXt [21]	51.67	78.40	19.38	42.61	66.04	92.05	58.36	82.16
	*GA-Nav [5]	<u>78.50</u>	88.25	37.28	72.34	<u>74.75</u>	<u>96.07</u>	<u>74.44</u>	91.69
	*EGFormer (ours)	79.70	<u>87.78</u>	38.12	<u>71.49</u>	75.83	96.16	74.85	<u>91.65</u>

TABLE II

STATISTICAL SIGNIFICANCE COMPARISON OVER THREE INDEPENDENT RUNS WITH DIFFERENT RANDOM SEEDS. WE REPORT MEAN $\pm$ STANDARD DEVIATION FOR mIoU AND bIoU ON RUGD AND RELIS-3D

Method	RUGD mIoU	RUGD bIoU	RELLIS mIoU	RELLIS bIoU
GA-Nav	88.94 $\pm$ 0.15	36.14 $\pm$ 0.91	73.90 $\pm$ 0.50	36.77 $\pm$ 0.42
EGFormer	89.18 $\pm$ 0.08	37.29 $\pm$ 0.07	74.82 $\pm$ 0.03	37.29 $\pm$ 0.04

that 0.5 provided the best trade-off between region-level segmentation and transition-focused supervision. This joint objective enables EGFormer to optimize both region-level accuracy and transition-region robustness, which is reflected in improved boundary-aware metrics in our experiments.

III. EXPERIMENTS

A. Experimental Setup

We evaluate EGFormer on two off-road semantic segmentation benchmarks, RUGD [22] and RELIS-3D [23]. Following GA-Nav [5] for fair comparison, all models are trained for 240 K iterations using SGD with polynomial learning-rate decay (power=0.9). Standard data augmentation (random cropping and horizontal flipping) is applied, and all methods are implemented using MMSegmentation [24]. The crop size and batch size are set to 300×375 and 24 for RUGD, and 375×600 and 6 for RELIS-3D, respectively. Unless otherwise specified, we set $\lambda_{GA} = 1.0$ and $\lambda_{edge} = 0.5$. We report mean IoU (mIoU), overall pixel accuracy (aAcc), and boundary IoU (bIoU) [25], [26], which measures performance near class transitions and is

particularly important for off-road drivability assessment; Table II reports mean $\pm$ standard deviation over three independent runs with different random seeds.

B. Main Results

Performance comparison: Table I compares EGFormer with prior CNN- and transformer-based semantic segmentation models on RUGD and RELIS-3D. EGFormer achieves competitive region-level performance while improving boundary-aware segmentation quality. On RUGD, EGFormer attains 89.25 mIoU and 95.78 aAcc, slightly improving over the GA-Nav baseline (89.08 mIoU), while on RELIS-3D it improves mIoU from 74.44 to 74.85. The gains are more pronounced in gradual or ambiguous transitions, where prediction errors are especially critical in off-road scenes. Table II further shows that EGFormer consistently outperforms GA-Nav across three random seeds on both datasets, while also exhibiting substantially lower variance (e.g., 0.03 vs. 0.50 on RELIS-3D mIoU and 0.04 vs. 0.42 on RELIS-3D bIoU), indicating more stable and reliable training behavior in ambiguous transition regions. The relatively larger variance of GA-Nav on RUGD bIoU (0.91) is likely attributable to annotation noise and ambiguous terrain transitions in RUGD, which make thin-boundary-focused predictions less stable across runs.

Qualitative results: Fig. 2 provides qualitative evidence under diverse boundary conditions. Compared with the baseline, EGFormer yields more consistent predictions in visually ambiguous transition regions and remains stable under label inconsistencies and annotation errors. In particular, transition boundaries are delineated more coherently without introducing

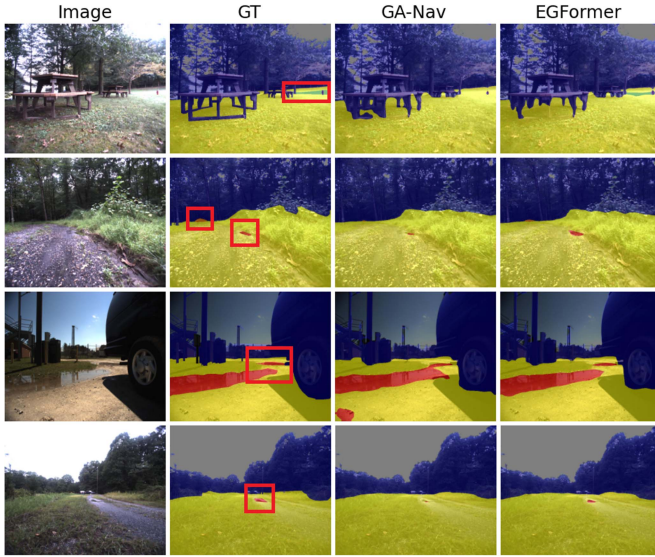


Fig. 2. **Unified qualitative comparison under diverse boundary conditions.** Columns show Image / GT / GA-Nav / EGFormer. EGFormer yields more stable and semantically consistent predictions across clear, ambiguous, and noisy boundary cases.

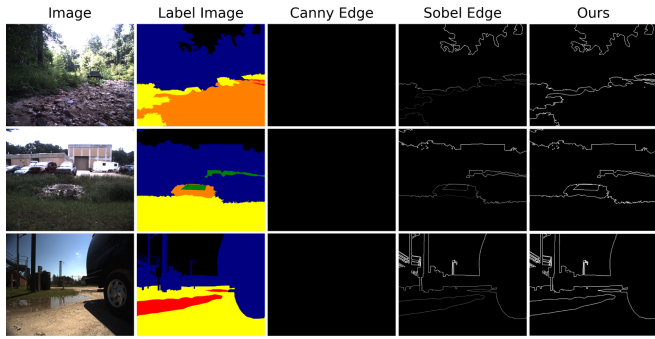


Fig. 3. **Comparison of transition-mask formulations.** From left to right: input image, group-label map, Canny edge, Sobel edge, and the proposed mask. The proposed mask better captures spatially extended transition bands.

artificially sharp edges, while isolated noisy responses are reduced and overall semantic structure is preserved. These qualitative improvements are especially visible near mixed terrain interfaces, where gradual appearance changes often make local predictions unstable. These results indicate that transition-focused refinement improves local reliability in mixed terrain regions without sacrificing global semantic consistency, consistent with the boundary-aware gains in Table II.

C. Ablation Study

Mask formulation comparison: Fig. 3 compares the proposed semantic-discontinuity mask with low-level edge cues obtained by Canny and Sobel operators. Unlike thin appearance-driven contours, the proposed formulation highlights semantically meaningful transition bands between terrain groups and more directly captures the gradual and ambiguous boundaries commonly observed in off-road scenes. This behavior, together with the quantitative comparison in Table III, indicates that the

TABLE III
IMPACT OF EDGE-POSITIVE WEIGHT α ON RELLIS-3D. A MODERATE VALUE ($\alpha = 16$) YIELDS THE BEST TRADE-OFF BETWEEN mIoU AND bIoU

Edge Positive Weight	mIoU	bIoU	Notes
-	74.44	37.12	GA-Nav
Dilated boundary band	74.46	37.17	3-pixel dilated band
$\alpha = 8$	74.40	37.25	Insufficient refinement
$\alpha = 16$	74.85	37.35	Optimal
$\alpha = 40$	74.43	37.25	Overfitting to edges

observed gain is not obtained by simply widening low-level boundaries, but by using transition-aware supervision tailored to semantically ambiguous off-road scenes.

Effect of edge-positive weight α : We analyze the influence of the edge-positive weight α in L_{edge} on RELLIS-3D (Table III). A moderate setting of $\alpha = 16$ achieves the best trade-off, improving both mIoU and bIoU. Smaller values under-emphasize transition supervision, while larger values bias learning toward edge regions and degrade stability. These results indicate that balanced weighting of transition-band pixels is important for robust boundary modeling.

Stability under ambiguous supervision: We examine whether edge refinement improves robustness rather than inducing overfitting. Across our experiments, EGFormer exhibits more stable predictions in transition regions under ambiguous supervision, consistent with the qualitative results in Fig. 2. This suggests that explicitly modeling transition bands helps preserve boundary coherence when supervision is imperfect. Moreover, the ER branch operates on fused decoder features and adds only modest overhead, increasing the number of parameters from 6.21 M to 6.40 M and the computational cost from 18.69 to 19.82 GFLOPs. This suggests that the observed gains arise from transition-focused refinement rather than from generic model scaling or a substantial increase in capacity.

IV. CONCLUSION

We presented EGFormer, a framework for off-road semantic segmentation that explicitly models semantically ambiguous terrain transition bands through a transition-aware refinement branch. Experiments on the RUGD [22] and RELLIS-3D [23] benchmarks demonstrate consistent gains in boundary-aware performance (bIoU), particularly in ambiguous transition regions, with minimal computational overhead. Ablation and qualitative analyses further show improved robustness under ambiguous supervision and noisy annotations. Future work will extend transition-aware supervision to more challenging sensing conditions and finer-grained boundary modeling.

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